

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: dataset = pd.read_csv("ONE PIECE.csv")
```

```
dataset.head()
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 958 entries, 0 to 957
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      958 non-null   int64
1   rank            958 non-null   object
2   trend           958 non-null   object
3   season          958 non-null   int64
4   episode         958 non-null   int64
5   name            958 non-null   object
6   start           958 non-null   int64
7   total_votes     958 non-null   object
8   average_rating  958 non-null   float64
dtypes: float64(1), int64(4), object(4)
memory usage: 67.5+ KB
```

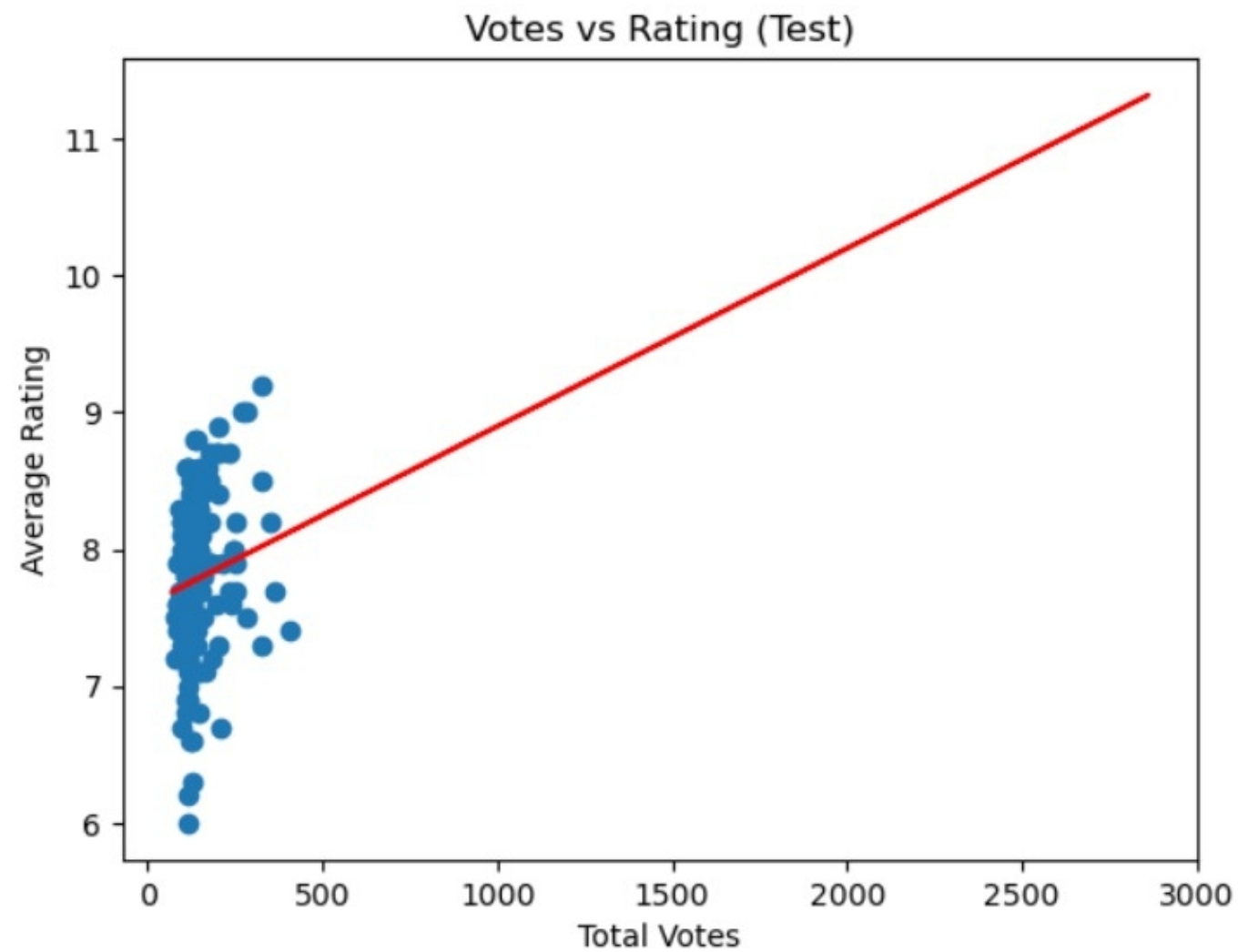
```
[3]: # Remove commas from total_votes
dataset['total_votes'] = dataset['total_votes'].astype(str).str.replace(',', '')
dataset['total_votes'] = pd.to_numeric(dataset['total_votes'])

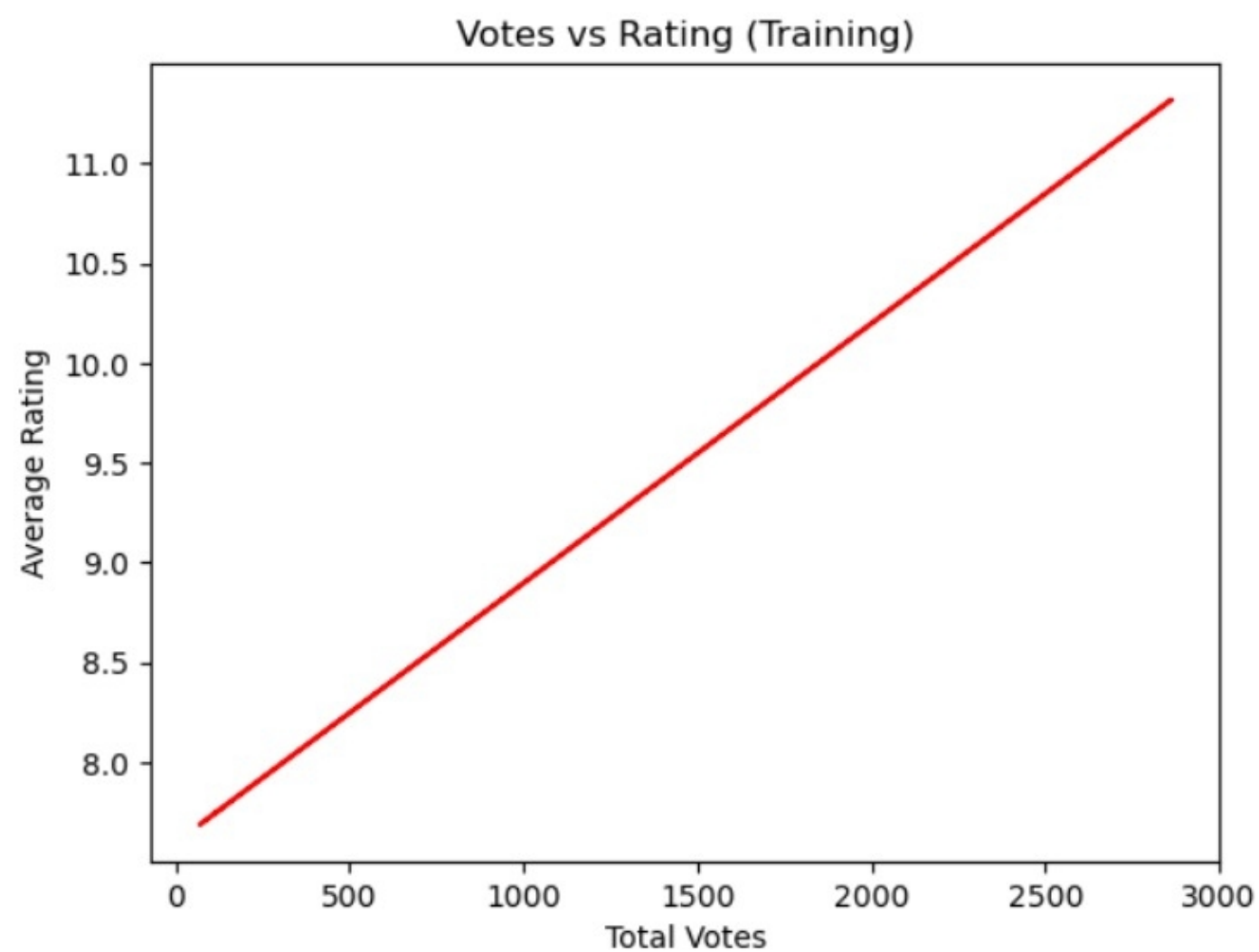
# Remove commas from rank
```

```
[11]: plt.scatter(X_test, y_test)
plt.plot(X_train, lr.predict(X_train), color='red')

plt.title("Votes vs Rating (Test)")
plt.xlabel("Total Votes")
plt.ylabel("Average Rating")

plt.show()
```





```
[11]: plt.scatter(X_test, y_test)
plt.plot(X_train, lr.predict(X_train), color='red')

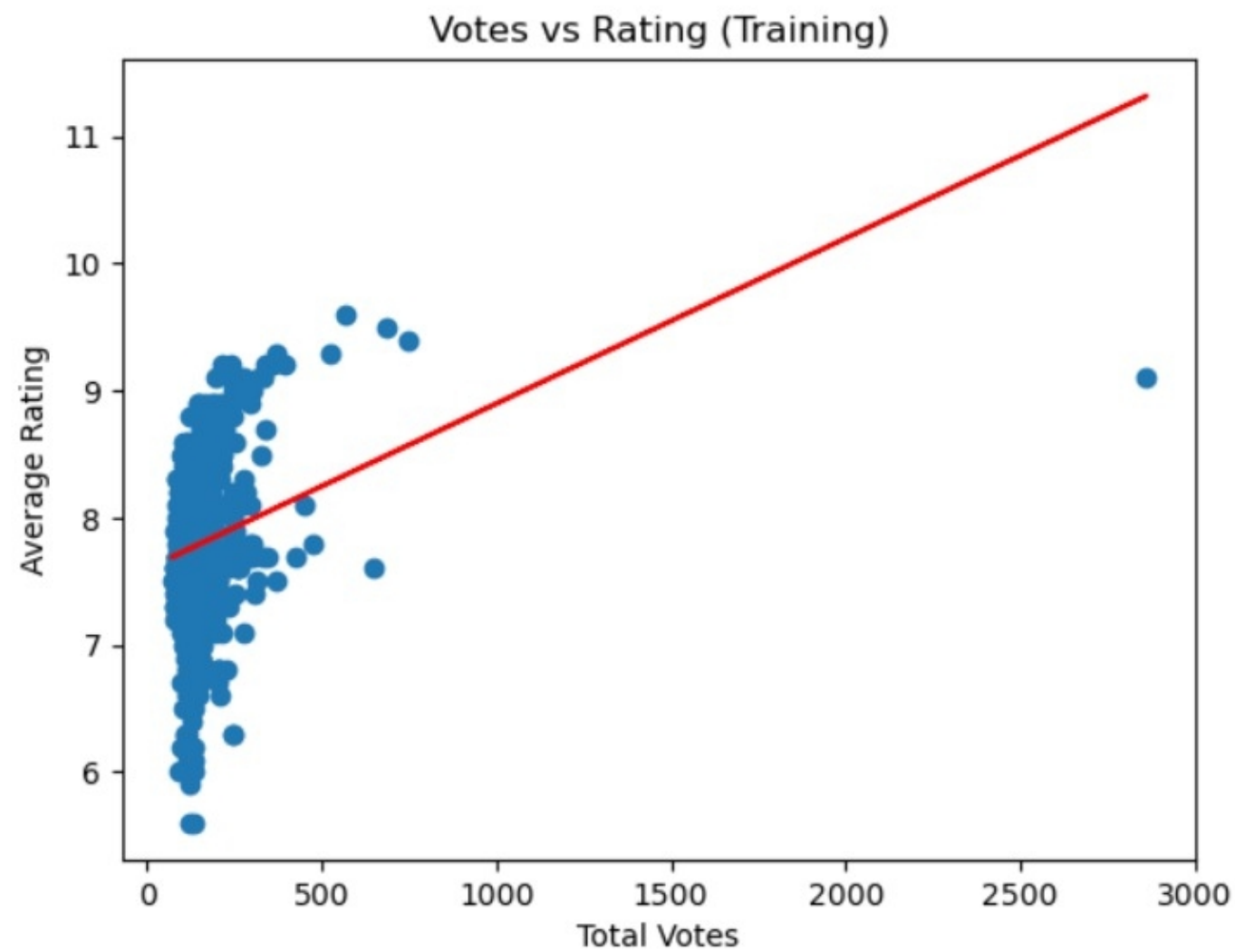
plt.title("Votes vs Rating (Test)")
plt.xlabel("Total Votes")
plt.ylabel("Average Rating")

plt.show()
```

```
plt.show()
plt.plot(X_train, lr.predict(X_train), color='red')

plt.title("Votes vs Rating (Training)")
plt.xlabel("Total Votes")
plt.ylabel("Average Rating")

plt.show()
```



```
[10]: plt.scatter(X_train, y_train)
plt.plot(X_train, lr.predict(X_train), color='red')

plt.title("Votes vs Rating (Training)")
plt.xlabel("Total Votes")
plt.ylabel("Average Rating")

plt.show()
plt.plot(X_train, lr.predict(X_train), color='red')

plt.title("Votes vs Rating (Training)")
plt.xlabel("Total Votes")
plt.ylabel("Average Rating")

plt.show()
```




```
[3]: # Remove commas from total_votes
dataset['total_votes'] = dataset['total_votes'].astype(str).str.replace(',', '')
dataset['total_votes'] = pd.to_numeric(dataset['total_votes'])

# Remove commas from rank
dataset['rank'] = dataset['rank'].astype(str).str.replace(',', '')
dataset['rank'] = pd.to_numeric(dataset['rank'])
```

```
[4]: X = dataset[['total_votes']].values
y = dataset['average_rating'].values
```

```
[5]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=0
)
```

```
[6]: from sklearn.linear_model import LinearRegression

lr = LinearRegression()
lr.fit(X_train, y_train)
```

```
[6]: LinearRegression ⓘ ?
      ▶ Parameters
```

```
[7]: y_pred = lr.predict(X_test)
```